INTERNAL HOM OF COHERENT COMPLEXES IS COHERENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-coherent-internal-hom; the proof environment immediately following it is its own proof

Editorial note. The label lemma-coherent-internal-hom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- membership in D_Coh is etale-local on a locally Noetherian algebraic space (spaces-cohomology-lemma-coherent-Noetherian of Cohomology of Spaces)
- the schemes case: for a scheme X, L in D^+_Coh and K in D^-_Coh implies RSheafHom(K, L) in D^+_Coh (perfect-lemma-coherent-internal-hom of Derived Categories of Schemes)

Source reference. Original lemma 0D0C: sources/stacks-chow/tags.txt, source lines 16307–16307. Prerequisite pointer invoked at proof line(s) 1469; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07UB: sources/stacks-chow/tags.txt, source lines 9966–9966. Prerequisite pointer invoked at proof line(s) 1467; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S. If L is in $D^+_{Coh}(\mathcal{O}_X)$ and K in $D^-_{Coh}(\mathcal{O}_X)$, then $R\mathcal{H}om(K, L)$ is in $D^+_{Coh}(\mathcal{O}_X)$.*

Proof. We can check whether an object of $D(\mathcal{O}_X)$ is in $D_{Coh}(\mathcal{O}_X)$ étale locally on X, see Cohomology of Spaces, Lemma 07UB. Hence this lemma follows from the case of schemes, see Derived Categories of Schemes, Lemma 0D0C. □

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